

1 DOF

1.1 变量

δ 容许弥散圆直径

f 焦距

F 光圈值

R 光圈半径

u, u_1, u_2 物距

v, v_1, v_2 像距

L, L_1, L_2 景深前景深后景深

1.2 基本等式

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{f} = \frac{1}{u_1} + \frac{1}{v_1}$$

$$\frac{1}{f} = \frac{1}{u_2} + \frac{1}{v_2}$$

$$R = \frac{f}{F}$$

1.3 v

$$v = \frac{fu}{u - f}$$

1.4 v_1

$$\frac{\delta}{R} = \frac{v_1 - v}{v}$$

$$v_1 - v = \frac{\delta}{R}v$$

$$v_1 = \frac{\delta}{R}v + v$$

$$v_1 = (1 + \frac{\delta}{R})v$$

1.5 u_1 L_1

$$u_1 = \frac{fv_1}{v_1 - f}$$

$$u_1 = \frac{f(1 + \frac{\delta}{R})v}{(1 + \frac{\delta}{R})v - f}$$

$$u_1 = \frac{f(1 + \frac{\delta F}{f})v}{(1 + \frac{\delta F}{f})v - f}$$

$$u_1 = \frac{f(1 + \frac{\delta F}{f})(\frac{fu}{u-f})}{(1 + \frac{\delta F}{f})(\frac{fu}{u-f}) - f}$$

$$u_1 = \frac{f(f + \delta F)(\frac{fu}{u-f})}{(f + \delta F)(\frac{fu}{u-f}) - f^2}$$

$$u_1 = \frac{f(f + \delta F)fu}{(f^2u + \delta Ffu) - f^2u + f^3}$$

$$u_1 = \frac{f^2u + \delta Ffu}{\delta Fu + f^2}$$

$$L_1 = u - u_1$$

$$L_1 = u - \frac{f^2u + \delta Ffu}{\delta Fu + f^2}$$

$$L_1 = \frac{(\delta Fu + f^2)u - f^2u - \delta Ffu}{\delta Fu + f^2}$$

$$L_1 = \frac{\delta Fu(u - f)}{f^2 + \delta Fu}$$

1.6 v_2

$$\frac{\delta}{R} = \frac{v - v_2}{v}$$

$$v - v_2 = \frac{\delta}{R}v$$

$$v_2 = (1 - \frac{\delta}{R})v$$

1.7 u_2 L_2

$$u_2 = \frac{fv_2}{v_2 - f}$$

$$u_2 = \frac{f(1 - \frac{\delta}{R})v}{(1 - \frac{\delta}{R})v - f}$$

$$u_2 = \frac{f(1 - \frac{\delta F}{f})v}{(1 - \frac{\delta F}{f})v - f}$$

$$u_2 = \frac{f(f - \delta F)v}{(f - \delta F)v - f^2}$$

$$u_2 = \frac{f(f - \delta F)(\frac{fu}{u-f})}{(f - \delta F)(\frac{fu}{u-f}) - f^2}$$

$$u_2 = \frac{f(f - \delta F)fu}{(f^2u - \delta Ffu) - f^2u + f^3}$$

$$u_2 = \frac{f(f - \delta F)fu}{f^3 - \delta Ffu}$$

$$u_2 = \frac{f^2u - \delta Ffu}{f^2 - \delta Fu}$$

$$L_2 = u_2 - u$$

$$L_2 = \frac{f^2u - \delta Ffu}{f^2 - \delta Fu} - u$$

$$\begin{aligned}
L_2 &= \frac{f^2 u - \delta F f u}{f^2 - \delta F u} - \frac{u(f^2 - \delta F u)}{f^2 - \delta F u} \\
L_2 &= \frac{f^2 u - \delta F f u - u f^2 + \delta F u^2}{f^2 - \delta F u} \\
L_2 &= \frac{\delta F u^2 - \delta F f u}{f^2 - \delta F u} \\
L_2 &= \frac{\delta F u(u - f)}{f^2 - \delta F u}
\end{aligned}$$

1.8 L

$$\begin{aligned}
L &= L_1 + L_2 \\
L &= \frac{\delta F u(u - f)}{f^2 + \delta F u} + \frac{\delta F u(u - f)}{f^2 - \delta F u} \\
L &= \delta F u(u - f) \left(\frac{1}{f^2 + \delta F u} + \frac{1}{f^2 - \delta F u} \right) \\
L &= \delta F u(u - f) \left(\frac{f^2 - \delta F u + f^2 + \delta F u}{f^4 - \delta^2 F^2 u^2} \right) \\
L &= \frac{2f^2 \delta F u(u - f)}{f^4 - \delta^2 F^2 u^2}
\end{aligned}$$

L 相关计算过程参考由于变量设定不同众说纷纭下列出相关参考推导

[2] [1]

参考文献

[1] unknown. 参 考2. <https://www.specim.com/>

how-to-calculate-the-depth-of-field-dof-with-the-specim-hyperspectral-cameras/

Accessed: 2025-11-08.

[2] zhihu. 景深概念与计算. <https://zhuanlan.zhihu.com/p/240711762>.

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